(3) the person who conducts the examination shall record the details of the examination and any reports from a dump person in the daily examination and report book referred to in section 6.4.2, and

(4) communicate verbally any abnormal or hazardous conditions to the dump person.

Shiftboss – Underground

Records

6.6.1 Every underground shiftboss shall record daily in the underground shiftboss's logbook any accumulation of flammable refuse and waste timber in the underground shiftboss's work area.

Fireboss – Underground Coal

Shift Inspections

6.7.1 (1) Within three hours before the beginning of each shift in an underground coal mine, a fireboss must

(a) inspect with a suitable direct reading atmospheric testing device that is used, calibrated and maintained in accordance with the manufacturer's instructions that part of the mine and the roadways leading to it through which persons may be present or pass, and

(b) make a report on the results of the fireboss's inspection.

(2) No person shall enter that part of the mine or the roadways referred to in subsection (1) until they have been inspected and declared safe by the fireboss.

Reports

6.7.2 A copy of the report required by 6.7.1 shall be signed by the fireboss, and a copy of it shall be posted at the surface of the mine.

Weekly Inspections

6.7.3 At least once in every week the manager shall cause a fireboss to examine all roadways, air courses, stoppings, sealings, overcasts, and wastes, with a suitable direct reading atmospheric testing device that is used, calibrated and maintained in accordance with the manufacturer's instructions, to make a report, and post a copy of it in accordance with section 6.7.2.

Mine Plans

Duty to Keep Plans Surface

6.8.1 The manager shall keep at the mine site accurate plans that are updated in accordance with good engineering practice and are prepared on a scale that accords with good engineering practice, as follows

(1) A surface plan showing the claims, licences or leases on which mining is being carried out, and all lakes, watercourses, naturally unstable ground, such as peat bogs or sloughs, main roads, railways, power transmission lines, buildings, shaft opening, adits, surface workings, diamond drill holes collared on the surface, dumps, dams, tailings ponds and their overflow channels, topographic contours, and any abandoned, adjacent or historical workings.

(2) The manager shall ensure that a thorough search has been completed for the existence of the abandoned, adjacent or historical workings, the results of this search shall be provided to an inspector and a copy maintained at the minesite for inspection.

Failure to Post Plans

6.8.2 If the manager fails to provide the plans required by section 6.8.1, the chief inspector may have the mine surveyed and the plans prepared, and the costs of the survey and the preparation of the plans may be recovered from the owner or agent.

Traffic Control

6.8.3 The manager shall prepare traffic control procedures, showing the maximum allowable speeds for the vehicles in use, rules for passing, "stop" and "yield rules," priority rules for various vehicles, rules for night operation, maximum operating grades, emergency run-off protection, shoulder barriers, and any other information that may be required to ensure the safe operation of all types of vehicles on the mine site.

Mine Haul Road Design

Haulage Road Width

6.9.1 The manager shall prepare a plan pursuant to section 10 (1) of the *Mines Act* which

(1) Shows the type and method of construction for haulage roads that are to be constructed at the mine site.

(2) Except for roads constructed prior to 1990, the manager shall ensure that haulage roads are designed, constructed and maintained to provide

(a) a travel width where dual lane traffic exists, of not less than 3 times, or where single lane traffic exists, of not less than 2 times the width of the widest haulage vehicle used on the road, and

(b) a shoulder barrier

(i) at least 3/4 of the height of the largest tire on any vehicle hauling on the road,

(ii) of a construction or a specification that is in general conformance to accepted engineering practice,

(iii) located and maintained along the edge of the haulage road wherever a drop-off greater than 3 m exists, and

(iv) incorporating breaks that do not exceed the width of the blade of the equipment constructing and maintaining the breaks to allow for drainage and snow clearance.

(3) For the purpose of subsection (2) (a), the width of the barrier referred to in subsection (2) (b) shall be excluded from the travel width.

Vehicle Runaway Protection

6.9.2 On roadways where the grade exceeds 5% the manager shall have installed and maintained runaway lanes or retardation barriers where conditions/risk warrant.

Dumps

Dumps, Roads and Ramps Manager's Responsibility

6.10.1 The manager shall require a qualified person to

(1) prepare and maintain a plan pursuant to section 10 (1) of the *Mines Act*, consistent with good engineering practice for dumps, stockpiles, minor impoundments, roads, or ramps that are to be constructed as part of a dumping operation, the plan shall include monitoring for safety,

(2) ensure that the construction is in accordance with the plan and any modification to the plan that has been approved by the manager,

(3) communicate the accepted plan and any modifications to those persons responsible for and employed in the construction,

(4) where material is to be dumped from a vehicle into a bin, raise, or other opening, provide and maintain a barrier of sufficient size and strength and anchored sufficiently to prevent the vehicle from inadvertently entering the bin, raise, or opening,

(5) where dumping is prohibited or hazardous, prevent dumping by placing across the entrance a barrier sufficient to prevent access and a sign that reads "No entry for dumping purposes,"

(6) appoint qualified persons to act as dump persons, and

(7) prepare a procedure for controlling access to areas within the potential run-out zone of all dumps; this procedure will prohibit extended activities below active dumps and provide for a program of monitoring to allow work below inactive and dormant dumps, the procedure will include provisions for signage, work under adverse conditions and shall be reviewed annually.

Respiratory Hazards

6.10.2 (1) The mine manager shall ensure that there are no confined spaces connected to or in close proximity of waste dumps.

(2) If the requirement of section 6.10.2(1) is impracticable, the mine manager shall have the structure, earthworks or culvert designed in accordance with established engineering practices such that no workers are at risk of exposure to deficient atmospheric oxygen.

Underground Mine Design Features

Boundary Pillar

6.11.1 A pillar of not less than 30 m shall be maintained on either side of a party boundary between adjoining underground mining properties.

6.11.2 The owners of adjoining properties may, by written agreement, waive the provisions of section 6.11.1, provided that they inform the chief inspector.

6.11.3 If the owners fail to reach an agreement under section 6.11.1 the chief inspector may, upon application from a manager, authorize the mining of a pillar.

Shaft Pillars

6.11.4 No stoping shall be done within 60 m of a shaft that is used for transporting persons, unless the mining plan has been prepared by a registered professional engineer and the mine manager has so authorized.

Mine Openings

Two Exits

6.12.1 Where an underground mine has been opened by means of an adit, tunnel, or shaft, a second or auxiliary exit shall be provided prior to production.

Distance Between Exits

6.12.2 An auxiliary exit shall not at any point be less than 30 m from the main entrance to underground workings of the mine and shall be separated by solid strata.

Auxiliary Exit

6.12.3 Before stoping is started in any part of a mine, a second or auxiliary exit shall be provided to the working place in that stoping block, or the mine manager shall ensure that a single exit would always be open.

Ease of Travel

6.12.4 An auxiliary exit shall afford easy passage and be provided with good and substantial ladders from the deepest working to the surface, or with hoisting apparatus, which shall constantly be available for use.

New Mine

6.12.5 Where an underground mine is being developed, shafts and raise openings to the surface shall be provided with a substantial collar secured to the bedrock, comprised of concrete, or other material that will perform in the same manner as concrete and is not expected to deteriorate.

Refuge Stations

Location of Refuge Station

6.13.1 Where a workplace in an underground mine is more than 300 m from a mine portal or from a shaft station which is used to access that workplace, the manager shall provide and maintain, in a suitable location for that workplace, a refuge station in accordance with section 6.13.3.

Exemptions

6.13.2 Section 6.13.1 does not apply to a mine under initial adit development or during shaft sinking operations.

Construction and Equipping

6.13.3 Every underground refuge station shall be

(1) clearly identified, constructed of non-combustible material, and of sufficient size to accommodate all persons working in the vicinity,

(2) equipped with a supply of air, a supply of water, a means of communicating with the surface, a means of sealing to prevent entry of gas, and first aid equipment,

(3) equipped with a plan of the mine clearly showing all emergency exits, and

(4) located

(a) 100 metres from explosives magazines,

(b) 100 metres from flammable materials storage, and

(c) constructed or located in such a manner to prevent inadvertent entrance or damage by vehicles, or

(5) In the case of an underground coal mine the manager will establish at appropriate locations storage facilities with suitable equipment to allow for emergency exit from the mine.

6.14.1 Repealed.

6.14.2 Repealed.

6.14.3 Repealed.

Track Haulage Systems

6.15.1 A track haulage system may be designated by the chief inspector as a railway and in that case, it shall be maintained and operated in accordance with the standards established under the *Railway Act*.

Rail Haulage Clearances

6.15.2 Subject to section 6.16.2, where underground track haulage is used the following continuous minimum clearances between the vehicle and the sides of the roadway shall be maintained

(1) 300 mm on one side and 600 mm on the other, and

(2) a minimum of 300 mm clearance shall be maintained above the head of a person traveling on the haulage vehicle and the roof of the roadway or any obstruction in it.

Underground Mobile Equipment

Mobile Equipment Clearances

6.16.1 Where any mobile equipment is operating, a minimum clearance of 2 m in excess of the maximum width of the vehicle shall be maintained. The vehicle shall be fitted with an approved roll-over-protective structure ROPS or falling object protection structure (FOPS) and there shall be maintained a minimum clearance of 300 mm above ROPS or FOPS.

Reduced Clearances

6.16.2 An inspector may permit reduction of the minimum clearances set out in section 6.15.2 and 6.16.1, if safety stations are provided at such distances and under such conditions as the inspector approves.

Safety Stations

6.17.1 The safety stations referred to in section 6.16.2 shall

(1) be plainly marked,

(2) be clean and free of obstructions,

(3) be cut as close to perpendicular as is practical to the haulageway, and

(4) if made after 1990, be at least

(a) 1 m in depth, in addition to the clearance between the vehicle and the wall,

(b) the height of the roadway or 2 m whichever is least, and

(c) 1.5 m in width.

Remote Control Operations and Emerging Technology

6.18.1 Before any equipment that can be moved by remote control is introduced at a mine, the manager shall

(1) Provide, and have approved by the chief inspector, a plan showing that the system, device or controller is capable of operating only the specific piece of equipment it is designed to operate.

(2) Ensure that other forms of energy are not capable of rendering the equipment inoperative causing uncontrolled activation or operation of the equipment.

(3) The controller be equipped with a lock-out device that renders it inoperative when not in use.

(4) Ensure the transmitter is equipped with an emergency stop mechanism that when activated applies the brakes and shuts down the equipment.

(5) For mobile equipment, if the transmitter is hand held, is equipped with a device that automatically works in the same manner as the emergency stop mechanism if the transmitter is tilted more than 15% from the level position.

(6) For fixed or tracked equipment a device which causes the machine to cease operating if controls are returned to the neutral position.

6.18.2 Where remote controlled equipment is to be used at a mine the manager shall have established operating procedures which shall include

(1) safe location for the operator,

(2) allows for a clear view of the working area,

(3) safe interaction between mechanical and remotely controlled equipment,

(4) a method for recovering equipment which has broken down, and

(5) if applicable, procedures to ensure contiguous operations do not interfere with remote signals.

Requirements for a Project Management Plan

6.18.3 (1) Prior to the use of an emerging technology, the manager must submit to the chief inspector a project management plan, prepared by a qualified professional, respecting the emerging technology.

(2) The project management plan must contain the following elements:

(a) a detailed risk assessment for the purpose of identifying, assessing and managing hazards;

(b) a summary of the safe working procedures in the Mine Health and Safety Program required under section 1.6.9 (1) (c.1);

(c) a summary of the project management plan, including project milestones and scope;

(d) details on location, access, infrastructure and equipment;

(e) a summary of key roles and responsibilities associated with the project;

(f) a summary of system functionality, redundancy, limitations and safety features;
(g) a commissioning plan;
(h) a maintenance and inspection plan;
(i) an operational plan;
(j) a description of the training program and competence assessment for working with and around the emerging technology as required in section 1.11.1 and 1.11.2 of this code;
(k) the process for investigating failures;
(l) a summary of the updates to the Mine Emergency Response Plan relating to the emerging technology as required under section 3.7.1 of this code;
(m) a gap assessment of this code to identify non-conformances and plans and timelines to address the non-conformances, prior to implementation of the project management plan;
(n) a summary of critical controls as identified in the risk assessment referred to in paragraph (a);
(o) if autonomous and semi-autonomous machines are operated on the mine site, an interaction plan for human-operated equipment and personnel;
(p) if battery or hybrid vehicles are operated on the mine site, a battery management plan;
(q) if hydrogen-powered vehicles are operated on the mine site, a hydrogen management plan.

(3) The manager must ensure the OHSC is provided with an opportunity to review and provide comments respecting the provisions of the project management plan that relate to the emerging technology and worker health and safety, as well as associated updates as required under subsection (5), prior to submission to the chief inspector.

(4) The manager must implement and adhere to the project management plan once the chief inspector confirms that it meets the requirements of this code and is appropriate for the mine site.

(5) If a material change is proposed to the operational use of an emerging technology, the manager must
(a) ensure that a qualified professional updates the project management plan,
(b) submit the updated plan to the chief inspector, and
(c) not make operational changes until the chief inspector confirms that the updated project management plan meets the requirements of this code.

Emerging Technology Certifications

6.18.4 The manager must ensure that emerging technology has been certified by an organization that is accredited by the Standards Council of Canada as meeting the requirements of

(a) an applicable CSA Standard,

(b) an applicable standard recognized by the Standards Council of Canada, if a standard referred to in paragraph (a) does not exist, or

(c) an applicable standard acceptable to the chief inspector, if neither standard referred to in paragraphs (a) and (b) exist.

Equipment Operation – General

Operator's responsibility

6.19.1 (1) The operator of a unit of mobile equipment

(a) is directly responsible for its safe operation and for maintaining full control of the unit and complying with the *Mines Act*, the regulations and this code insofar as the operation of the equipment is concerned, and

(b) must wear a seatbelt and drive with the headlights and, where required, a flag equipped whip antenna light or a flashing light on at all times.

(2) A person who enters commands or inputs information into an autonomous or semi-autonomous system that governs the behavior of tracked or rubber-tired mobile equipment, must do so in a manner that ensures the safe operation of the equipment and that the system can maintain full control of the mobile equipment.

Logbook to be Maintained

6.19.2 The manager must provide and maintain an effective logbook system for each unit of mobile equipment over 7000 kg gross vehicle weight in which the vehicle operator prior to operating of the vehicle, or the qualified person examining the equipment under 6.19.3 (2),

(a) notes the condition,

(b) notes any unsafe conditions, and

(c) in the case of a qualified person making repairs, notes any repairs made and if the vehicle is safe to operate.

Examination of Equipment

6.19.3 (1) The operator of equipment must examine and check the equipment and read the logbook before putting the equipment into use and, if an unsafe condition is discovered, the operator must not operate the equipment until

(a) repairs have been made, or

(b) a qualified person has assured the operator that it is safe to operate the equipment and noted the reason in the logbook.

(2) In the case of autonomous or semi-autonomous tracked or rubber-tired mobile equipment, the manager must
   (a) when the unit is serviced, or at a minimum of every 24 operating hours, ensure that a qualified person examines and checks the equipment and reads the logbook, and
   (b) if an unsafe condition is discovered, ensure that the equipment is not put into operation until
      (i) repairs have been made, or
      (ii) a qualified person has assured the manager that the equipment is safe to put into operation and noted the reason in the logbook.

Name of Person

6.19.4 Every notation made in the logbook, maintenance record, and every other record relating to the condition of the equipment shall show the time and date of the entry and the name of the person who made the entry.

Restricted Vision

6.19.5 No person shall operate a unit of mobile equipment where the person's field of vision is restricted unless the person
   (1) has inspected the area into which the equipment is to be moved and, without delay, proceeds to operate the equipment, or
   (2) is directed by a signal person who is located in a safe position and in continuous contact with the operator of the equipment, or
   (3) is directed by a traffic control or warning system.

Disabled Vehicle

6.19.6 When a vehicle or a unit of mobile equipment is disabled or parked in the traveled portion of a roadway
   (1) a warning to approaching traffic shall be given by means of flashing lights, flares, lamps, or reflectors, or
   (2) a person, equipped to be clearly visible and identifiable, shall direct other vehicles using that section of roadway.

Haulage Vehicle Operations – Surface

Fly Spreading

6.20.1 Where it is proposed to discharge material from the elevated box of a moving haulage truck, the manager shall prepare a safe operating procedure.

Dump Stability

6.20.2 No person shall drive or operate a haulage vehicle in such a manner as to

(1) dump from the vehicle at a location where the person has reasonable cause to believe that the ground is not capable of supporting the loaded vehicle, or

(2) remove material from the bottom of a dump or stockpile if the person has reasonable cause to believe that a person may be injured as a result of the removal.

Dumping Over Bank

6.20.3 No person shall drive or operate a haulage vehicle, in such a manner as to dump material from the vehicle over a bank that is more than 3 m high, or dump within 3 m of the dump berm crest when the bank is more than 3 m high, except as described in section 6.10.1(4), unless a dump person is directing vehicles to the dumping position and a dump berm is in place.

Reverse When Dumping

6.20.4 The driver of a haulage truck shall not

(1) where the bank is more than 3 m high and the dumping position is within 3 m of the dump berm crest, move the vehicle backward to the dumping position or begin dumping until the driver has received directions from the dump person,

(2) operate the vehicle in reverse for a distance greater than 4 truck lengths on a dump other than a bin, raise, or other opening referred to in section 6.10.1 (4), or

(3) operate the vehicle in reverse for a distance greater than 4 truck lengths, on a stockpile, ramp, road, or a ramp or road that is under construction, unless the ramp or road has a positive gradient of more than 5% or the procedure is accepted as part of a permit application or work system approval.

Dump Person

6.20.5 A dump person who is responsible for directing vehicles at a dump point shall

(1) continually inspect the condition of the dump site and if abnormal or hazardous conditions are observed take corrective action to alleviate any danger to workers assigned to the dump, and

(2) communicate immediately any abnormal or hazardous conditions found to the open pit shiftboss or, in the case of a mine with fewer than 6 employees, the supervisor.

Battery Charging

Battery Charging Stations

6.21.1 A battery charging station in a mine or mine shop shall be located in a well-ventilated location to prevent the accumulation of flammable gases.

Crew Transportation – Rail Track

Riding on Cars

6.22.1 No person, unless authorized by the manager, shall ride on or against a car, locomotive, or train in an underground working.

Passenger Cars

6.22.2 Section 6.22.1 does not apply where persons are being transported on or in passenger cars especially provided for that purpose in compliance with 3.2.4 and hauled by mechanical means provided that

(1) if the car does not have a roof, there shall be a clearance of not less than 1.1 m above each seat,

(2) if the car does have a roof, there shall be a clearance of not less than 150 mm above the roof, and

(3) safety chains shall be used between the cars and between the first car and the locomotive, in addition to the normal couplings except where an insulated draw bar is required.

Surface Mines

Removal of Unconsolidated Material

6.23.1 All trees and other vegetation, clay, earth, sand, gravel, loose rock, or other unconsolidated material lying within 2 m of the rim of a working face or wall in a surface mine shall be removed, and beyond this distance all unconsolidated material shall be sloped to an angle less than the natural angle of repose.

Bench and Berm Widths

6.23.2 Where a surface mine is worked in benches

(1) each catchment berm shall be designed so that its final width will not be less than 8 m,

(2) notwithstanding section 10.5.8, loose rock and soil shall not be allowed to accumulate on a bench or catchment berm in a manner that endangers any person working on a lower bench, and

(3) where loose rock accumulates and where access cannot be gained to clean the catchment berm, and a danger exists to a person working below, a safe working procedure shall be developed.

Surface Mine Faces and Bench Heights

6.23.3 No part of a face or wall of a surface mine shall overhang.

6.23.4 Where material is being worked or removed, the vertical component of the mining face shall not be greater than 2 meters beyond the reach of the loading equipment.

6.23.5 Section 6.23.4 does not apply

(1) where material is removed by backhoe, excavator, dragline or similar equipment operating from above the face that it is excavating, or

(2) where a multiple bench system of mining is being carried on in accordance with conditions authorized by the **chief permitting officer**, or

(3) where the material is free running and the slope does not exceed 60 degrees or 30 meters in length.

General Mine Rules

6.24.1 Surface excavations shall be securely fenced against inadvertent access when fencing is considered necessary by an inspector.

6.24.2 Wherever practicable, water sprays or other dust suppression means and devices shall be used at every dusty place where work is carried out and where it is impractical to do so, personal protective equipment shall be supplied and worn by all persons working in that location.

Drilling

6.24.3 A powered rock drill shall not be used in a mine unless it is equipped with a water jet or other device capable of suppressing dust.

General – Underground Rules

Examination

6.25.1 (1) Before any work is begun in an underground mine, a worker shall check the worker's workplace for hazardous or dangerous conditions and the worker shall not start work until the workplace has been made safe.

(2) Adit entrances and all other openings to underground mines that are no longer in use shall be secured against entry in a manner acceptable to an inspector.

6.25.2 The manager shall ensure that all work places, travel ways and other areas that may be frequented by mine personnel are regularly inspected and maintained by check scaling as conditions warrant and re-supported as may be prudent and that adequate tools and supplies are provided.

Guarding Open Workings

6.25.3 Where persons are working in any location which is below and open to a haulage or travel way on the level above, the workplace shall be securely covered or otherwise closed off from the haulage or travelway.

Guarding Millholes and Passageways

(1) The top of every millhole, passageway, sump, drain hole or other opening shall be covered or otherwise adequately protected to prevent inadvertent access.

(2) Where repair work is being carried out in a passageway, or where for any other reason the passageway could be dangerous to a person entering it, the passageway shall be closed off and warning signs shall be posted at all entrances.

Old Abandoned Workings

6.25.4 No work shall be carried out within 30 m of abandoned or old workings, or any accumulation of water or unconsolidated material, or any other substance that may flow, unless the proposed work procedure has been approved by the manager.

Breakthrough to Mine Workings

6.25.5 No connection between mine workings shall be made until a thorough examination of the workings toward which the active heading is advancing has been made and has shown that the work can proceed in a safe manner.

Jumbo Drills

6.25.6 No person shall proceed beyond the front of the drill controls of an underground jumbo drill unless the drills have been stopped.

Rock Passes

Water Not To Be Introduced

6.26.1 No person shall introduce water into an ore or waste pass, a loading pocket, coal silo, hopper, storage bin or a completed large diameter borehole for any purpose without following an operating procedure approved by the chief inspector.

Procedure for Removal

6.26.2 If any ore or waste pass is found to contain water, or saturated material which will flow, the manager shall provide a plan for the safe removal of such water or saturated material.

6.26.3 The controls of any ore or waste pass shall be arranged, wherever practicable, to prevent the operator from being hit by a run of material.

Stairways

Requirements

6.27.1 Stairways shall be provided in shafts or passageways inclined at 50 degrees from the horizontal or less where necessary to ensure safety.

Handrail

6.27.2 Every stairway greater than 1.5 m in height located in a shaft or passageway shall be equipped with a suitably placed handrail.

Tread Size

6.27.3 The rise and tread width of the steps within a stairway shall be uniform and tread widths shall not be less than 250 mm.

Guardrails

6.27.4 When a stairway ends in direct proximity to dangerous traffic or other hazards, detour guardrails shall be installed.

Ladders

General Standards

6.28.1 A ladder used in a mine shall

(1) be of strong construction,

(2) where used underground, be securely fastened to the timbering or wall of the shaft, raise, or stope

(3) be maintained in good repair,

(4) have rungs that are equally spaced,

(5) be installed so that rungs shall be more than 100 mm from the wall or timbering of a shaft, raise or stope, and

(6) be erected at an inclination of not more than 80 degrees from the horizontal unless equipped with safety hoops.

Handrails

6.28.2 Every ladder must project at least 1 m above its landing platform, except where

(1) strong suitable handrails are provided on the platform, and

(2) the ladder is securely fastened to its platform.

Platforms

6.28.3 Where a shaft or passageway, driven after the proclamation of this code, is inclined at over 50 degrees from the horizontal, it shall be provided with a ladderway containing platforms erected at vertical intervals not exceeding 7.5 m in the ladderway. Each platform shall have an opening which permits a person wearing a self- contained breathing apparatus to pass through.

6.28.4 Where a shaft or passageway is inclined at over 70 degrees from the horizontal, the ladderway, in addition to the requirements of section 6.28.3, shall have the individual ladders between each platform so located that they act as shields or protections to cover the holes in the platforms.

Doors in Platforms

6.28.5 Where it is possible for a person to fall from one platform to a lower platform in a ladderway, the platform openings shall be covered by suitable doors which can be easily opened from above or below.

Ladderways in Other Mine Workings

6.28.6 A landing platform shall be installed at all points where ladders are offset.

6.28.7 All ladderways in raises, stopes and passageways shall be installed and maintained to reduce the hazard of a person falling.

Travel in Passageways with Skips

6.28.8 When a ladderway and a skipway occupy the same compartment in a passageway

(1) the ladders shall be adequately protected from material being hoisted,

(2) no person shall travel or be allowed in the ladderway while the skip is in motion except to handle material in the skip, and

(3) sufficient guardrails shall be installed to prevent a person using the ladderway from falling into the skipway.

Flexible Ladders

6.28.9 No person shall use, or allow to be used, a flexible ladder except

(1) in a shaft sinking operation,

(2) in an emergency rescue operation, and

(3) provided that the ladder is safe and secure.

Raise – Underground

Two- Compartment Raises

6.29.1 Except where approved raise driving equipment is used, every raise inclined at more than 50 degrees from the horizontal and being driven more than 16 m slope distance, shall:

(1) be divided into at least two compartments, one of which shall be maintained as a ladderway and equipped with suitable ladders maintained to within 12 m of the face as it is being driven, and

(2) if the second compartment is used to hold rock excavated from the face, it shall either be kept full, or suitable protection shall be provided to prevent a person from falling into this compartment.

Flammable Waste & Scrap – Underground

Removal

6.30.1 All waste, including waste timber, shall be removed from an underground mine on a regular basis and shall not be piled up nor permitted to decay in the mine.

Storage and Removal

6.30.2 In an underground mine, or in or about a headframe or shaft-house, flammable refuse shall be

(1) deposited in covered, fire-resistant containers, and

(2) removed at least weekly from the mine, headframe, or shaft-house.

Sulphide Dust

6.31.1 In any underground mine where the sulphur content of the ore is high and a sulphide dust explosion has occurred, the manager shall approve and implement a scheme for minimizing the danger from a sulphide dust explosion including

(1) provision for ensuring that all persons are removed to a place of safety prior to blasting, taking into consideration the mine layout and the ventilation circuits, and if necessary, removed to the surface of the mine,

(2) provision for ensuring that all accessible headings, raises, and other workplaces within 30 m of a blasting site are washed down to remove dust prior to blasting, and

(3) any other measures that could reduce the risk of a dust explosion or lessen the consequences if one should occur.

Fire Doors – Underground

Fire Doors

6.32.1 In any mine, there shall be a sufficient number of fire doors installed underground to isolate the shaft or other entrance to the mine from the mine workings.

6.32.2 Fire doors shall be

(1) of fire resistant construction and of at least one-hour fire resistance rating,

(2) maintained in proper order,

(3) kept clear of all obstructions so as to be readily usable at all times, and

(4) equipped with a self-closing door to allow the escape of a person trapped behind it.

Ventilation Standard

6.33.1 The return air from all working places shall, whenever practicable, be routed directly to the return airway.

Barometer

6.33.2 A barometer and a thermometer shall be placed in a conspicuous position at the entrance of an underground coal mine, and both shall be read as part of the duties of the persons appointed to examine the mine workings under Sections 6.7.1 and 6.7.3.

Main Ventilation – Interruptions

Interruption to Main Fan

6.34.1 If the main system of ventilation for an underground mine is stopped, other than through a brief interruption of the power supply, all persons shall be withdrawn to the surface of the mine or to an approved refuge station in accordance with the manager's emergency procedures, and there shall be no entry of persons until the ventilation has been restored and the workings inspected and declared safe by an authorized person.

Underground Mine Heating

Mine Air Heaters

6.35.1 No furnace or device for heating mine air shall be installed without the written permission of an inspector.

Auxiliary Ventilation

Provisions of Ventilation

6.36.1 The manager shall ensure that all workings that are removed from the main ventilation circuit have an adequate supply of ventilating air.

Manager's Rules

6.36.2 Where auxiliary ventilation systems are necessary, the manager of an underground mine shall prepare rules and procedures for the installation and use of auxiliary ventilation systems and, in the case of a coal mine, for the degassing of headings. A copy of the procedures shall be posted at a conspicuous location at the mine.

No Recirculation of Air

6.36.3 An auxiliary fan shall not be installed or operated in an underground mine other than in accordance with the rules and procedures prepared under section 6.36.2, and the manager shall ensure that

(1) sufficient fresh air reaches the fan at all times to prevent re-circulation,

(2) air circulated by the fan is not contaminated with dust or flammable gas in excess of the limits specified in the rules and procedures prepared under section 6.3 6.2, and that no air containing more than 1.25% by volume of flammable gas passes through or over the fan,

(3) every auxiliary fan shall be electrically connected to ground to prevent the accumulation of an electrostatic charge,

Aluminum Parts Prohibited

(4) no auxiliary or booster fan shall be installed or operated in an underground coal mine if any of its component parts is made of aluminum, magnesium, titanium or a light metal alloy unless adequately coated with a non-sparking material, and

(5) the manager shall establish an inspection program to regularly inspect coated fans and shall have them immediately removed from service if the coating is damaged.

Compressed Air Machine Exhaust

6.36.4 The exhaust from any compressed air machine shall not be considered as ventilation.

Ventilation Monitoring

Air Measurements

6.37.1 The manager of an underground mine shall ensure that measurements of the quantity of air flowing are taken at the main fan, in all main airways, and at all major ventilation splits at intervals not exceeding 3 months.

Diesel Equipment Ventilation

6.37.2 In a mine or part of a mine in which diesel equipment is operating underground

(1) measurements of the quantity of air flowing shall be taken at intervals not exceeding once a week,

(2) tests shall be made at least once a shift, in the general body of the air, on the exhaust side of the operating diesel equipment, for nitrogen dioxide or oxides of nitrogen and other gases specified by an inspector, and

(3) Repealed.

(4) the worker may request that tests be conducted to determine the volume of air flow, carbon monoxide, nitrogen dioxide, formaldehyde, or diesel particulate matter contents of the atmosphere.

Recording

6.37.3 The measurements required to be taken under sections 6.37.2(1) and 6.37.2(2) shall be recorded in a book kept for that purpose.

Doors and Regulators

Door Closing

6.38.1 All doors used for regulating or controlling ventilation shall close automatically and be provided with a suitable device for opening or closing them.

Quantities – Splits – Measurement for Coal Mines

Ventilation Splits

6.39.1 Where necessary for efficient and effective ventilation, every mine shall be divided into separate ventilating splits and no working place shall be ventilated by series ventilation.

Measurements

6.39.2 Air measurements shall be taken to determine the quantity of air circulating

(1) within 90 m from the first normal working place in each split intake,

(2) in the intake and return airways of each ventilating split, and

(3) in the main intake and return airways of the mine.

Recording

6.39.3 The measurements referred to in section 6.39.2 shall be taken once per month and the results shall be entered in a book to be kept at the mine.

Unventilated Workings

Prevention Of Access

6.40.1 The manager shall ensure that any part of an underground mine that is not being ventilated shall

(1) be effectively barricaded to prevent inadvertent entry,

(2) be posted with signs warning that entry is prohibited, and

(3) before anyone enters or is permitted to enter that part of the mine, be examined by a qualified person for

(a) oxygen deficiency,

(b) the presence of toxic or noxious gases, fumes, mists, vapours, or dust, and

(c) any other dangerous condition.

6.40.2 All worked out or closed parts of an underground coal mine and those not part of the mine ventilation system shall be sealed with substantial stoppings.

Stoppings – Coal Mines

Stopping Construction

6.41.1 All stoppings between intake and return airways shall be of substantial construction and built in a manner to prevent any undue leakage of air.

6.41.2 The space between the faces of all stoppings and the airways shall be kept free of obstructions.

Monitoring

6.41.3 Provisions shall be made for monitoring the conditions behind stoppings.

Flammable Gas

Gas Detection

6.42.1 Where it may be reasonable to expect that there is a hazard due to the presence of flammable gas, an inspector may require that a sufficient number of appliances of an approved type be provided for the determination of the percentages of flammable gas.

Qualifications

6.42.2 The manager of an underground mine shall appoint a qualified person to make regular determinations of the content of flammable gas in the mine air at such locations and at such time intervals as established by the manager. The results of the determinations shall be entered in a book kept for this purpose at the mine.

Reporting

6.42.3 If air immediately returning from a split that ventilates a group of active workings, or if air in the main return airway, is found to contain more than 1% of flammable gas as determined with an approved means of detection, the person who detects it shall immediately report the person's findings to the manager and the manager shall

(1) take immediate steps to improve the ventilation, and

(2) send a notice of the condition to an inspector within 24 hours.

Electrical Equipment

6.42.4 If any person finds more than 1.25% of flammable gas in the air in the general vicinity of electrical machinery or equipment the person shall cut off the electrical power supply from the machinery or equipment and report the circumstances to the supervisor in charge of that part of the mine.

Coal Mines – Withdrawal Of Persons

6.42.5 (1) Whenever the general body of the air in any workplace or travelway in a mine is found to contain 2.5% or more of flammable gas, the person in charge of that part of the mine shall

(a) immediately cause all persons in that area and the return airway to be withdrawn to a place of safety,

(b) inform without delay the person's immediate supervisor, and

(c) insofar as it is possible to do so without undue risk, ascertain or have a qualified person ascertain, the condition of the affected area and the return airway and carry out the measures necessary to render it safe

Examination

(2) Except for the purpose of saving life or rendering the area safe, no person shall be readmitted to the affected area until the person in charge of that part of the mine, or a qualified person appointed by the manager, has examined the area and reported it to be safe.

Recording

(3) Where persons have been ordered to leave an affected area under this section, the person who caused them to leave shall record in a book kept for that purpose at the mine, particulars of the reason for the order, and the person who carried out the examination required in subsection 6.42.5(2) shall report the result of it in that book

Prevention of Ignition

Manager's Responsibility

6.43.1 Where a machine is in use at a working face at which gas is liable to be ignited by such use, the manager shall ensure that

(1) the machine is provided with effective means to prevent the ignition, or

(2) an adequate portable fire extinguisher is available at or near the machine.

6.43.2 The manager shall provide a detector that gives an audible and/or visible warning of a hazardous accumulation of flammable gas where electrically powered machinery is used at the coal face or at any other place specified by an inspector.

Prohibited Metals

6.43.3 Nothing made of, or containing, aluminum, magnesium, titanium, or light metal alloy shall be used in an underground coal mine, except

(1) electrical equipment within a flameproof enclosure, or

(2) in circumstances when there is no possibility of friction or impact, or

(3) unless adequately coated with non-sparking material and immediately removed from service if the coating is damaged, or

(4) hand held tools which are placed in a non-sparking storage container following use.

Automatic Gas Detector

6.43.4 Where electrically powered equipment is being operated at a coal face in an underground mine, the manager shall ensure that an automatic gas detector of a type that is approved by a recognized body is installed. The detector shall be installed in such a manner that if a predetermined level of flammable gas is detected, all electrically powered machinery is shut down and is not restarted until gas levels have fallen below the prescribed level.

Gas Detectors

6.43.5 (1) The supervisors shall check all gas detectors in underground coal mines daily, by comparing the reading obtained from the fixed detector with the reading obtained from a supervisor's detector.

(2) If a variance of 10% or more occurs in daily readings, both detectors are to be immediately removed from service, replaced before operations restart, and calibrated against a known concentration of gas.

(3) All gas detectors in underground coal mines shall be removed once per month, serviced and tested against a known concentration of flammable gas.

(4) The manager shall ensure that records are kept of all calibration tests and maintenance carried out on gas detection devices and that those records are available to an inspector on request.

Combustible Dust – Coal Mines

Treatment

6.44.1 The floor, roof and sides of every road or part of a road that is accessible shall be treated

(1) with water in the manner and at the intervals that will ensure that the dust on the floor, roof, and sides, respectively, is always combined throughout with 30% by weight of water in the intimate mixture, or

(2) with incombustible dust in a manner and at intervals that will ensure the dust on the floor, roof, and sides, respectively, shall always consist throughout of a mixture containing not more than 50% of combustible matter if the volatile matter content of the coal does not exceed 22% as determined by one of the standard methods of analysis and computed on a dry, ash free basis.

Method

(3) Before the first application of incombustible dust as required by this part of the code, accumulated coal dust shall be removed from the roof, floor and sides of the roadway, so far as practicable.

Permissible Levels

6.44.2 The maximum permissible percentage of combustible matter under section 6.44 1 shall diminish by 1.5% for each 1% increase of volatile matter of the coal until it has been reduced to 35 in the case of coal having a volatile matter content of 32% or more.

Samples

6.44.3 For the purpose of determining the volatile matter content of the coal under section 6.44 2, samples shall be taken either from representative sections of the seam or from a representative quantity of the run of mine coal from the seam.

Methane

6.44.4 The permissible percentage of combustible matter in the dust found in an underground roadway shall be further decreased by one for each increment of 1/10 part of 1% in the methane content of the mine air beyond 1/4 of 1%.

Moisture

6.44.5 The percentage of incombustible dust required under this section may be reduced by an amount equivalent to the percentage of water present in the mixture.

Exception

6.44.6 The obligations imposed by this section do not apply to a roadway, if the natural conditions of it as regards the presence of incombustible dust and moisture are found by tests made in accordance with this section to be such as to comply with the foregoing requirements.

Fineness of Dust

6.44.7 The incombustible dust used for the purpose of this part of the code shall, whenever possible, contain not less than 50% by weight of fine material capable, when dry, of passing through a No. 80 sieve of the Canadian Metric Sieve Series.

Sampling

6.44.8 If the amount of incombustible dust passing through the No. 80 sieve is less than 50%, the percentage of combustible matter specified as being the maximum permissible by sections 6.44.1(2), 6.44.2, 6.44.4 and 6.44.5 shall be decreased proportionately, but the percentage of the fine material shall never fall below 25%.

6.44.9 To obtain the composition of the dust mixture in a road or part of a road, the following procedure shall be adopted

(1) representative samples of the dust shall be collected from the floor, roof, and sides over a section of the road not less than 30 m in length, the sections being not more than 135 m apart in the same roadway, and

(2) each sample collected shall be thoroughly mixed and quartered, and a portion of the mixture shall then be sifted through a No. 315 sieve of the Canadian Metric Sieve Series

Analysis of Roadway Dust

6.44.10 If the roadway dust is known to contain only negligible percentages of either gypsum or carbonates

(1) a weighed quantity of the dust that has passed through the sieve shall be dried at a temperature of 105 degrees Celsius and the weight lost shall be reckoned as moisture, and

(2) the sample shall then be brought to a red heat in an open vessel until it no longer loses weight, and the weight lost by incineration shall be reckoned as combustible matter for the purpose of the test.

Gypsum

6.44.11 If the incombustible dust applied to a roadway consists of gypsum wholly or in part

(1) a weighed quantity of the sieved dust shall be dried at a temperature between 135 degrees Celsius and 140 degrees Celsius and the weight lost shall be reckoned as moisture, and

(2) the sample shall then be kept at a red heat in an open vessel until complete incineration, and the weight of the residue added to that of the moisture shall be reckoned as incombustible matter and expressed as a percentage of the total weight of sieved dust treated.

Carbonate

6.44.12 If the roadway dust contains an appreciable proportion of carbonates, the following method shall be followed

(1) a weighed quantity of the dust that has passed through the sieve shall be dried at a temperature of 105 degrees Celsius, and one hundred times the weight lost divided by the number of grams of dust submitted to the test shall be reckoned as the percentage of moisture, and

(2) one gram of the sample so dried shall then be treated with dilute hydrochloric acid in a suitable apparatus; the weight lost through the decomposition of the carbonates shall be ascertained and subsequently added to that of the incombustible solid residue of